

SHUYU LEI

PhD Candidate Msc, BEng Urban Planning

Transport & Urban Planning | Inclusive Mobility | Climate Resilience

The University of Hong Kong, Hong Kong SAR | leishuyu@connect.hku.hk | Urban design Portfolio | ORCID

About Me

Transport and urban planning specialist translating mobility, spatial, and climate data into evidence for inclusive policy and investment. Peer-reviewed research spans age-friendly accessibility, extreme heat and vulnerable transit users, thermal comfort in transit-oriented development, and spatial-network links to multidimensional poverty in Nairobi.

Qualifications

PhD, Urban Planning

The University of Hong Kong
Transport studies specialization

Expected Feb 2027

MSc, Urban Planning

The University of Hong Kong
Royal Town Planning Institute & Hong Kong Institute of Planners
student member

2019-2021

Exchange, Urban Planning: Cities, Environment & Landscapes

Politecnico di Milano, Italy
China Scholarship Council award

2018

BEng, Urban & Rural Planning

Shenzhen University, China
Outstanding Graduate award

2014-2019

Skills

Methods: Mixed-method research design; qualitative interviews and field observation; survey design and analysis; statistical modelling and confounder control; spatial accessibility and network analysis.

Tools: Python; R; ArcGIS Pro; QGIS; SPSS; Rhino, sketchup, photoshop etc. autocad.

Communication: Data visualization; cartography; report and presentation design; Illustrator; InDesign

Languages

Mandarin Chinese (Native) | English (Full professional proficiency) | Cantonese (Full professional proficiency) | Japanese (Elementary) | Italian (Elementary)

Policy Writing & Academic Service

Contributing Author

Intermodal Transport Data-Sharing Programme: Final Report, Innovation and Technology Fund project.

2021

Lead Contributor

Green Travel City Toolbook: Shared Public Space (2024); Green Travel City Toolbook: Inclusive Mobility (2025). 9.22 Green Travel Platform, supported by the China Academy of Urban Planning and Design and the Urban Planning Society of China.

2024-2025

Book Manuscript Contributor, forthcoming

Two chapters of Policy Analysis for Integrated Urban Transport Governance: Concepts and Methods, developed in collaboration with Tongji University; currently in copyediting.

Ad Hoc Journal Reviewer, 2026

Journal of Transport & Health; Sustainable Cities and Society; Journal of Transport Geography

Professional Experience

Doctoral Researcher – Urban and Transport Planning

Feb 2023-Present | The University of Hong Kong | Hong Kong SAR

Accessibility: Built an age-sensitive X-minute-city framework from 35,401 households, 101,385 individuals, 66,740 facility locations, four age cohorts, and a 3D pedestrian network.

Transit analytics: Processed one month of smart-card data covering approximately 5.9 million passenger IDs across 90 stations and 11 lines; applied clustering, PCA, MANOVA, spatial analysis, and confounder control.

Policy translation: Published two first-author journal articles and produced station-level recommendations for staff assistance, wayfinding, barrier-free routes, and priority interventions.

Senior Research Assistant - Life between Stations

Nov 2025-Present | The University of Hong Kong | RGC General Research Fund | Hong Kong SAR

Contribution: Process and visualize mixed-method data on station pairs, active-mode links, spatiotemporal patterns, and rider wellbeing to identify first-/last-mile interventions.

Research Assistant - Smart Travel Services

May 2021-Nov 2023 | HKU-Shenzhen Institute of Research and Innovation | Shenzhen / Hong Kong SAR

Regional infrastructure: Analysed travel behaviour and supported cross-boundary mobility forecasting and information services for the Hong Kong-Zhuhai-Macao Bridge.

Research Assistant - Intermodal Data-Sharing

Jul 2020-Feb 2021 | The University of Hong Kong | Innovation and Technology Fund GSP/016/19 | Hong Kong SAR

Seven-city policy analysis: Benchmarked integrated transport, data governance, demand-responsive mobility, engagement, and decarbonization across Copenhagen, London, Paris, Sydney, Seoul, Shenzhen, and Chicago; evidence supported four report chapters circulated to 600+ stakeholders.

Urban Planning Intern

Aug-Sep 2018 | Yongchun County Planning Project | Fujian, China

Government engagement: Conducted field investigation and presented village renewal and master-planning proposals to county stakeholders.

Applied Research & Projects

Spatial Networks & Multidimensional Poverty

Published 2025 | Applied Geography | Taita Taveta University collaboration | Nairobi, Kenya

Development evidence: Linked 756 geocoded households with 16 network indicators; higher density and efficiency were associated with lower poverty, while severance increased risk.

City Connect 15+ - GIS Accessibility Tool

2023 | CSDI Awards | Merit Award, Open Category | Hong Kong SAR

Public-facing tool: Co-developed an age-sensitive 15-minute-access prototype covering approximately 137,000 grid cells, eight facility categories, 2SFCA, a 3D pedestrian network, barrier-free access, and weather scenarios.

Selected Policy-Relevant Research

POVERTY & SPATIAL EQUITY | KENYA

Association between multidimensional poverty and urban spatial network design: Comparison between theory-driven and data-driven lenses

Gachanja, J., **Lei, S.**, Adero, N. | 2025 | *Applied Geography* 178, 103578

Policy relevance: Identifies neighbourhood-scale network priorities and highlights informal-settlement conditions in poverty interventions.

CLIMATE RESILIENCE | VULNERABLE GROUPS

Disparities in the impacts of extreme heat events on subway ridership in New York City

Zhao, M., **Lei, S.**, Li, S., Li, W. | 2026 | *npj Sustainable Mobility and Transport* 3, 17

Policy relevance: Shows the largest ridership declines among seniors and passengers with disabilities, especially on weekends.

CLIMATE-RESILIENT TOD

Incorporating urban thermal comfort into transit-oriented development planning: Non-linear heterogeneous built environment effects

Zhao, M., **Lei, S.**, Li, W. | 2026 | *Sustainable Cities and Society* 140, 107254

Policy relevance: Links greater building-height variation and vegetation ratios with improved station-area thermal comfort.

INCLUSIVE MOBILITY | OLDER ADULTS

Rethinking the X-minute city for older adults: Temporal thresholds, hierarchical needs and age-specific walking accessibility

Lei, S., Zhao, M., Zhou, J., Yang, X. | 2026 | *Journal of Transport Geography* 134, 104656

Policy relevance: Finds older adults' typical walking thresholds fall around 5-12 minutes and maps priority low-accessibility areas.

INCLUSIVE PUBLIC TRANSPORT

Evaluating age-friendly public transportation: A novel smart card data approach analyzing travel time differences between older and younger travelers after controlling for confounding effects

Lei, S., Zhou, M., Deng, W., Tang, Y., Zhou, J. | 2026 | *Transportation Research Part A* 204, 104842

Policy relevance: Provides a scalable diagnostic for complex interchanges and stations where older passengers experience greater travel-time penalties.

Presentations & Teaching

CONFERENCE PRESENTATION

Oral presenter: TRB 103rd Annual Meeting, Washington, DC (Jan 2024), Towards Age-Friendly Station Design, TRBAM-24-02874

Additional Publications & Report

RouteKG: A knowledge graph-based framework for route prediction on road networks

Tang, Y., Zhao, Z., Deng, W., **Lei, S.**, Liang, Y., Ma, Z. | 2025 | *IEEE T-ITS* 26(12), 22277-22295

Intentional travel group and social network: Identification and dynamics during a pandemic

Zhou, M., **Lei, S.**, Wu, J., Ma, H., Levinson, D. M., Zhou, J. | 2025 | *Environment and Planning B* 52(4), 823-840

'Right here waiting'? Distribution of intentional metro travel groups amid COVID-19 in Hong Kong

Zhou, M., **Lei, S.**, Zhou, J. | 2024 | *Transactions in Urban Data, Science, and Technology* 3(3-4), 136-139

Towards niche market for shared mobility: Identifying heterogeneity of potential early adopters to use shared electric bikes in a Chinese mega city

Wu, J., **Lei, S.**, Zhou, J., Wang, X., Tao, Z. | 2023 | *Transactions in Urban Data, Science, and Technology* 2(4), 241-256

Introducing social contacts into the node-place model: A case study of Hong Kong

Zhou, M., Zhou, Jiali, Zhou, Jiangping, **Lei, S.**, Zhao, Z. | 2023 | *Journal of Transport Geography* 107, 103532

Age-friendly public transport: Evidence from travel behaviors of older adults in Hong Kong's metro station

Lei, S., Zhou, Jiali, Zhou, Jiangping | 2023 | *Smart City Symposium Prague*, 1-8

Population density, activity centres, and pandemic: Visualizing clusters of COVID-19 cases in Hong Kong

Zhou, Jiangping, Ho, S. K. S., **Lei, S.**, Pang, V. C. K. | 2021 | *Environment and Planning A* 53(7), 1594-1597

Intermodal Transport Data-Sharing Programme: Final Report

2021 | *The University of Hong Kong* | Global Cities Research Assistant: Lei Shuyu

INVITED KNOWLEDGE SHARING

Invited speaker: Harbin Institute of Technology, Shenzhen International School of Design (Dec 2024), Mind the Gap: From Research to Age-Friendly Design

Invited guest lectures: Shanghai University of Finance and Economics (4 Jul 2024); Tongji University (11 Jun 2024); Southwestern University of Finance and Economics (2023)